1 UK Constraints Derivation

We will simulate two six degree of freedom bodies joined by a revolute joint. Let the first object be object 1, and the second object be object 2. Each object has a position and an attitude. For each object, the state is comprised of the inertial position and the inertial to body orientation represented as MRPs.

$$x_i = \begin{bmatrix} r^0_{i/0} \\ \sigma^i_0 \end{bmatrix} \tag{1}$$

The overall state being the concatenation of these two states:

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} r^0_{1/0} \\ \sigma^1_0 \\ r^0_{2/0} \\ \sigma^2_0 \end{bmatrix} \tag{2}$$

There are two parts of the constraint, positional and rotational. Let $r^1_{j/1}$ represent the position of the joint on body one in the body frame, and j_1 be the direction vector of the joint on body one in the body frame.

The position and rotation constraints are:

$$\phi_r = r^0_{1/0} + C((\sigma^1_0)^{-1})r^1_{j/1} - r^0_{2/0} - C((\sigma^2_0)^{-1})r^2_{j/2} \tag{3}$$
$$\phi_\sigma = (C((\sigma^1_0)^{-1})j_1) \times (C((\sigma^2_0)^{-1})j_2) \tag{4}$$

Where $C(\sigma)$ is the DCM derived from the MRPs (using the following formula):

$$C(\sigma) = I - \frac{4(1-\sigma^T\sigma)}{(1+\sigma^T\sigma)^2}\tilde{\sigma} + \frac{8}{(1+\sigma^T\sigma)^2}\tilde{\sigma}^2 \tag{5}$$

With these two constraints defined, we can take the second derivative to get the constraint equations $A\ddot{x} = b$.

$$\phi_r = r^0_{1/0} + C((\sigma^1_0)^{-1})r^1_{j/1} - r^0_{2/0} - C((\sigma^2_0)^{-1})r^2_{j/2} \tag{6}$$

$$\begin{aligned} \frac{d\phi_r}{dt} &= \dot{r}^0_{1/0} + \dot{C}((\sigma^1_0)^{-1})r^1_{j/1} - \dot{r}^0_{2/0} - \dot{C}((\sigma^2_0)^{-1})r^2_{j/2} \\ &= \dot{r}^0_{1/0} + C((\sigma^1_0)^{-1})\tilde{\omega}^1_{1/0}r^1_{j/1} - \dot{r}^0_{2/0} - C((\sigma^2_0)^{-1})\tilde{\omega}^2_{2/0}r^2_{j/2} \end{aligned} \tag{7}$$

$$\begin{aligned} \frac{d^2\phi_r}{dt^2} &= \ddot{r}^0_{1/0} + C((\sigma^1_0)^{-1})\tilde{\alpha}^1_{1/0}r^1_{j/1} + C((\sigma^1_0)^{-1})(\tilde{\omega}^1_{1/0})^2 r^1_{j/1} \\ &\quad - \ddot{r}^0_{2/0} - C((\sigma^2_0)^{-1})\tilde{\alpha}^2_{2/0}r^2_{j/2} - C((\sigma^2_0)^{-1})(\tilde{\omega}^2_{2/0})^2 r^2_{j/2} \end{aligned} \tag{8}$$

$$\begin{aligned} \frac{d^2\phi_r}{dt^2} &= \ddot{r}^0_{1/0} - C((\sigma^1_0)^{-1})\tilde{r}^1_{j/1}\alpha^1_{1/0} + C((\sigma^1_0)^{-1})(\tilde{\omega}^1_{1/0})^2 r^1_{j/1} \\ &\quad - \ddot{r}^0_{2/0} + C((\sigma^2_0)^{-1})\tilde{r}^2_{j/2}\alpha^2_{2/0} - C((\sigma^2_0)^{-1})(\tilde{\omega}^2_{2/0})^2 r^2_{j/2} \end{aligned} \tag{9}$$

Before stepping into ϕ_σ, let us define some terms that will be useful, and take their derivatives:

$$
\begin{aligned}
u &= C((\sigma_0^1)^{-1})j_1 && (10)\\
\dot{u} &= C((\sigma_0^1)^{-1})\tilde{\omega}_{1/0}^1 j_1 && (11)\\
\ddot{u} &= C((\sigma_0^1)^{-1})\tilde{\alpha}_{1/0}^1 j_1 + C((\sigma_0^1)^{-1})(\tilde{\omega}_{1/0}^1)^2 j_1 \\
&= -C((\sigma_0^1)^{-1})\tilde{j}_1\alpha_{1/0}^1 + C((\sigma_0^1)^{-1})(\tilde{\omega}_{1/0}^1)^2 j_1 && (12)\\
v &= C((\sigma_0^2)^{-1})j_2 && (13)\\
\dot{v} &= C((\sigma_0^2)^{-1})\tilde{\omega}_{2/0}^2 j_2 && (14)\\
\ddot{v} &= C((\sigma_0^2)^{-1})\tilde{\alpha}_{2/0}^2 j_2 + C((\sigma_0^2)^{-1})(\tilde{\omega}_{2/0}^2)^2 j_2 \\
&= -C((\sigma_0^2)^{-1})\tilde{j}_2\alpha_{2/0}^2 + C((\sigma_0^2)^{-1})(\tilde{\omega}_{2/0}^2)^2 j_2 && (15)
\end{aligned}
$$

$$
\begin{aligned}
\phi_\sigma &= u \times v && (16)\\
\frac{d\phi_\sigma}{dt} &= \dot{u} \times v + u \times \dot{v} \\
&= (C((\sigma_0^1)^{-1})\tilde{\omega}_{1/0}^1 j_1) \times v + u \times (C((\sigma_0^2)^{-1})\tilde{\omega}_{2/0}^2 j_2) && (17)\\
\frac{d^2\phi_\sigma}{dt^2} &= \ddot{u} \times v + 2\dot{u} \times \dot{v} + u \times \ddot{v} \\
&= (-C((\sigma_0^1)^{-1})\tilde{j}_1\alpha_{1/0}^1 + C((\sigma_0^1)^{-1})(\tilde{\omega}_{1/0}^1)^2 j_1) \times v \\
&+ 2(C((\sigma_0^1)^{-1})\tilde{\omega}_{1/0}^1 j_1) \times (C((\sigma_0^2)^{-1})\tilde{\omega}_{2/0}^2 j_2) \\
&+ u \times (-C((\sigma_0^2)^{-1})\tilde{j}_2\alpha_{2/0}^2 + C((\sigma_0^2)^{-1})(\tilde{\omega}_{2/0}^2)^2 j_2) && (18)
\end{aligned}
$$

In their current form, we cannot use these expressions, as we need to get rid of the angular acceleration terms (and replace them with $\ddot{\sigma}$). We can start with the dynamics equation for MRPs relating $\dot{\sigma}$ to ω, and differentiate it.

$$
\begin{aligned}
\omega &= 4B^{-1}\dot{\sigma} = 4\frac{(1-\sigma^T\sigma)I - 2\tilde{\sigma} + 2\sigma\sigma^T}{(1+\sigma^T\sigma)^2}\dot{\sigma} && (19)\\
\alpha &= 4B^{-1}\ddot{\sigma} + 4\dot{B}^{-1}\dot{\sigma} && (20)\\
\dot{B}^{-1} &= \frac{(-2\sigma^T\dot{\sigma}I - 2\tilde{\dot{\sigma}} + 2\dot{\sigma}\sigma^T + 2\sigma\dot{\sigma}^T)(1+\sigma^T\sigma) - 4\sigma^T\dot{\sigma}((1-\sigma^T\sigma)I - 2\tilde{\sigma} + 2\sigma\sigma^T)}{(1+\sigma^T\sigma)^3} && (21)
\end{aligned}
$$

Plugging Equation (20) into Equations (9) and (18), we get:

$$
\begin{aligned}
\frac{d^2\phi_r}{dt^2} &= \ddot{r}_{1/0}^0 - C((\sigma_0^1)^{-1})\tilde{r}_{j/1}^1(4B(\sigma_0^1)^{-1}\ddot{\sigma}_0^1 + 4(\dot{B}(\sigma_0^1))^{-1}\dot{\sigma}_0^1) + C((\sigma_0^1)^{-1})(\tilde{\omega}_{1/0}^1)^2 r_{j/1}^1 \\
&- \ddot{r}_{2/0}^0 + C((\sigma_0^2)^{-1})\tilde{r}_{j/2}^2(4B(\sigma_0^2)^{-1}\ddot{\sigma}_0^2 + 4(\dot{B}(\sigma_0^2))^{-1}\dot{\sigma}_0^2) - C((\sigma_0^2)^{-1})(\tilde{\omega}_{2/0}^2)^2 r_{j/2}^2 && (22)\\
\frac{d^2\phi_\sigma}{dt^2} &= (-C((\sigma_0^1)^{-1})\tilde{j}_1(4B(\sigma_0^1)^{-1}\ddot{\sigma}_0^1 + 4(\dot{B}(\sigma_0^1))^{-1}\dot{\sigma}_0^1) + C((\sigma_0^1)^{-1})(\tilde{\omega}_{1/0}^1)^2 j_1) \times v \\
&+ 2(C((\sigma_0^1)^{-1})\tilde{\omega}_{1/0}^1 j_1) \times (C((\sigma_0^2)^{-1})\tilde{\omega}_{2/0}^2 j_2) \\
&+ u \times (-C((\sigma_0^2)^{-1})\tilde{j}_2(4B(\sigma_0^2)^{-1}\ddot{\sigma}_0^2 + 4(\dot{B}(\sigma_0^2))^{-1}\dot{\sigma}_0^2) + C((\sigma_0^2)^{-1})(\tilde{\omega}_{2/0}^2)^2 j_2) && (23)
\end{aligned}
$$

Using these, we can get A and b for our UK constraint equation

$$
A = \begin{bmatrix} I_3 & -4C((\sigma_0^1)^{-1})\tilde{r}_{j/1}^1 B(\sigma_0^1)^{-1} & -I_3 & 4C((\sigma_0^2)^{-1})\tilde{r}_{j/2}^2 B(\sigma_0^2)^{-1} \\ 0_{3\times3} & 4\tilde{v}C((\sigma_0^1)^{-1})\tilde{j}_1 B(\sigma_0^1)^{-1} & 0_{3\times3} & -4\tilde{u}C((\sigma_0^2)^{-1})\tilde{j}_2 B(\sigma_0^2)^{-1} \end{bmatrix} \quad (24)
$$

$$
\begin{aligned}
b_r &= 4C((\sigma_0^1)^{-1})\tilde{r}^1_{j/1}(\dot{B}(\sigma_0^1))^{-1}\dot{\sigma}_0^1 - C((\sigma_0^1)^{-1})(\tilde{\omega}^1_{1/0})^2 r^1_{j/1} \\
&\quad - 4C((\sigma_0^2)^{-1})\tilde{r}^2_{j/2}(\dot{B}(\sigma_0^2))^{-1}\dot{\sigma}_0^2 + C((\sigma_0^2)^{-1})(\tilde{\omega}^2_{2/0})^2 r^2_{j/2}
\end{aligned} \tag{25}
$$

$$
\begin{aligned}
b_\sigma &= -4\tilde{v}C((\sigma_0^1)^{-1})\tilde{j}_1(\dot{B}(\sigma_0^1))^{-1}\dot{\sigma}_0^1 + \tilde{v}C((\sigma_0^1)^{-1})(\tilde{\omega}^1_{1/0})^2 j_1 \\
&\quad - 2(C((\sigma_0^1)^{-1})\tilde{\omega}^1_{1/0} j_1) \times (C((\sigma_0^2)^{-1})\tilde{\omega}^2_{2/0} j_2) \\
&\quad + 4\tilde{u}C((\sigma_0^2)^{-1})\tilde{j}_2(\dot{B}(\sigma_0^2))^{-1}\dot{\sigma}_0^2 - \tilde{u}C((\sigma_0^2)^{-1})(\tilde{\omega}^2_{2/0})^2 j_2
\end{aligned} \tag{26}
$$